

Distributional effects of vehicle ownership restrictions

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Overview

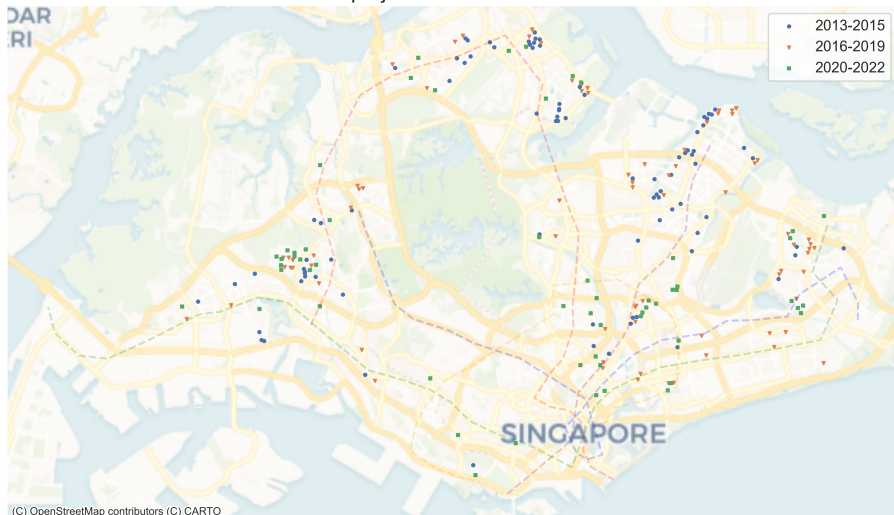
- 1 Introduction
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Traffic congestion is costly

- Traffic congestion is costly!
 - Cost the US economy nearly \$87 billion in 2018 (Fleming 2019)
 - 2019: Each Manilaño spent 10 days and 17 hours stuck in traffic (Agbisit 2021)
- How?
 - Lost productivity and pollution
 - may **impact disadvantaged communities** more than others
 - and have **differential impacts across time**
- In response, city governments **limit car registrations** and use
- The **design** of these policies is of first order importance

Disparate impacts by geography

BTO projects from 2013 to 2022



What are the **distributional impacts** of vehicle ownership restrictions, and how do these impacts **vary across time?**

- ① Describe our **setting**: the Vehicle Quota System in Singapore
 - in which registrations of new vehicles are limited by permits (“COEs”);
 - allocated at auction
 - similar to systems in Shanghai, Guangzhou, etc. (e.g., Xiao, Zhou, and Hu 2017; Li 2018)
 - and feature 10-year-long cycles (peak - trough: \$100,000)
- ② **Theoretical example** generates **herding** in car replacement
- ③ **Simple calibrated model** replicates COE price trends
 - Limited data: Aggregates in GDP growth, population
 - Can give replacement rates, gov. revenues in 2024-2025
- ④ **Full model** (following Gillingham et al 2022) harnesses microdata
 - to model heterogeneous consumers
 - thus enriching COE dynamics
 - allowing us to speak to efficiency and distributional concerns.

Setting: Vehicle Quota System in Singapore

In Singapore, new vehicle registrations are limited by quota.

- Every new car must be accompanied by a **Certificate of Entitlement** (COE, “permit”) valid for ten years
 - allocated at fortnightly auctions
 - in one of five categories (small car, big car, goods vehicles, motorcycles, “open”)
 - **renewable** once for five or ten years at the prevailing permit price (scrap value not refunded)
- When sold as a used car: permit “follows” the car
- Permit supply arises only from deregistrations of scrapped cars
 - Scrapped? Car owner refunded $f(\text{price paid for permit})$

Setting: The government's stated objective

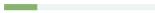
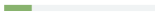
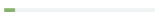
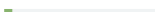
In Singapore: ministries set direction; statutory boards execute

- Ministry of Transport (MOT) problem: maximise social welfare(?)
- Land Transport Authority (LTA) problem:

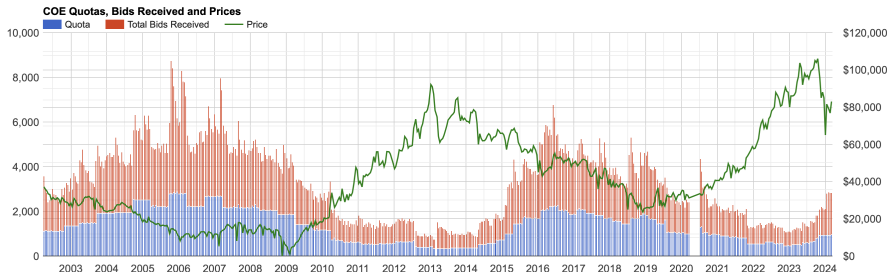
max revenue s.t. zero car growth

Zero car growth since 2018.

COE revenues are huge! (4% of Singapore's budget)

	Total revenue (\$)	% of total	% change 2023 
TOTAL REVENUE	132.1b	100%	 +3.9%
Revenue	132.1b	100%	 +3.9%
Corporate tax	28b 	21	 -1
NIRC	23.5b 	18	 +3
GST	19.4b 	15	 +19
Personal tax	18.1b 	14	 +3
Other taxes	8.86b 	7	 +0.9
Assets taxes	6.67b 	5	 +13
Stamp duty	5.73b 	4	 -3
Vehicle quota premiums	4.73b 	4	 +1
Other fees & charges	4.25b 	3	 +4
Customs, excise & carbon	3.56b 	3	 +5
Betting tax	3.26b 	2	 +3
Vehicle taxes	2.84b 	2	 +9

COE prices fluctuate wildly over time (\$100k over 10 years)



Do the negative externalities from car ownership fluctuate by \$10k/year?

Why model demand for COEs?

Modeling COE demand enables us to

- Compute tax incidence and welfare by car owner group.
- Evaluate policies, such as bringing forward COE supply when low.
- Inform 7th largest gov. revenue source (FY2024: S\$4.7b of S\$108.6b)

Conditional on existing household types (by residence location, etc),

- Input: LTA's COE policy + other vehicular/driving taxes.
- Output:
 - Household decisions over car purchase/replacement.
 - (Simultaneously) COE supply and prices over time.
 - (And thus) gov. revenue, welfare over time.

Recovering consumer preferences from a dynamic model seems difficult.
We leverage recent advancements in Industrial Organisation to dramatically simplify the estimation.

Theoretical Example

A model of two-period COEs generates “herding”

- LTA says owners “rush to replace” their cars when prices fall. Can our model replicate this descriptive fact?

Consumers (measure 1), One type of car (“free”), one class of COEs (measure 1/2, price p), valid 2 periods. Assume

- Consumer flow utilities from car ownership: $u_i \sim \text{Uniform}[0, 1]$
- Consumers are myopic
- “Old” COEs worth half of initial value
- Non-owners buy/not; Owners scrap/hold/trade

Non-owner and owner problems

Non-owners decide to buy or not. Buy iff

$$v_0(\text{buy}, u) = u - p \geq 0 = v_0(\text{not buy}, u)$$

Owner problem: Let p_{-1} be last period's COE price. Owner chooses one of the following three actions:

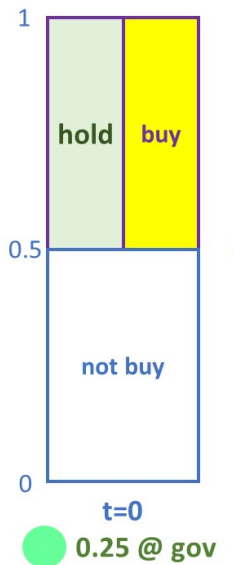
- Scrap: $v_0(\text{scrap}, u) = 0.5 \times p_{-1}$
- Hold: $v_0(\text{hold}, u) = u$
- Trade: $v_0(\text{trade}, u) = 0.5 \times p_{-1} - p + u$

Can verify the following steady state in “period 0”:

- COE price: $p_0 = 1/2$
- (mass 0.5) low-value consumers ($u \in [0, 1/2]$) never try for a car
- (mass 0.25) high-value consumers ($u \in [1/2, 1]$) have an old car
- (mass 0.25) high-value consumers have no car and will try for one

No one sells off their car early!

Visual proof: $p_0 = 0.5$ generates a steady state



Negative utility shock incentivizes early renewal

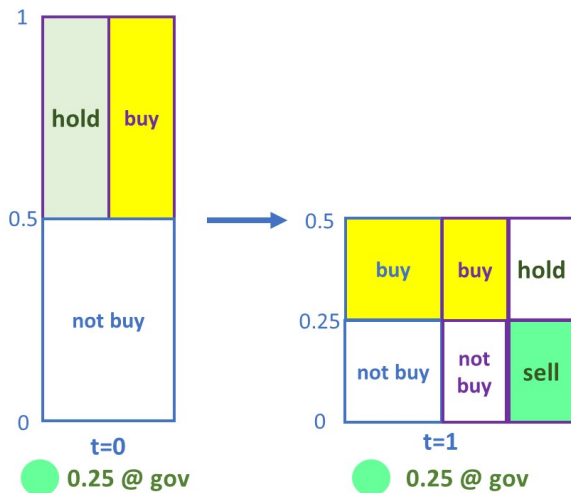
In period 1, suppose everyone with high valuations $u_i \in [1/2, 1]$ draws new valuations distributed iid $u'_i \sim \text{Uniform}[0, 1/2]$.

- e.g., negative income shock to rich, COVID shock...

Guess new COE price in first transitional period: $p_1 = 0.25$

- Non-owners buy car iff $u'_i \geq p_1$
- Owner values:
 - $v_1(\text{scrap}, u) = 0.5 \times p_0 = 0.25$
 - $v_1(\text{hold}, u) = u$
 - $v_1(\text{trade}, u) = 0.5 \times p_0 - p_1 + u = 0.25 - p_1 + u$

Visual proof: $p_1 = 0.25$ clears the market



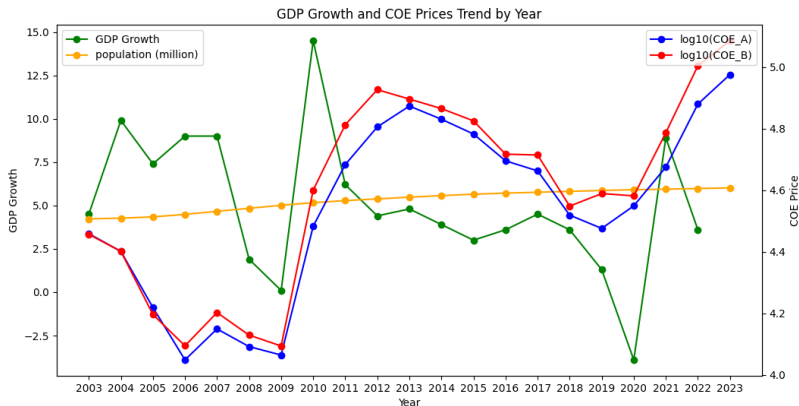
Our theoretical example replicates “herding”

In this new equilibrium,

- a mass $0.125 > 0$ of agents sells off their cars early.
- **Replicates “herding” behaviour when COE prices fall.**

Suggestive Calibration

Key Factors? Income (GDP Growth) & Population



- GDP growth is correlated with COE prices.

Suggestive Calibration

- In advance of LTA microdata [▶ Requested Data](#), calibrate a ten-period version of the simple model using 2009-2022 aggregates, e.g., population, GDP (proxying for income)
- Consumer flow utilities from car ownership are the product of a persistent component ζ_i and a function of the aggregate state s_t :

$$u_{it} = \zeta_i \times h(s_t), \quad \zeta_i \sim iid U[0, 1].$$

- Aggregate state s_t relates to lags of GDP growth and the population of Singapore:

$$h(s_t) = c_0 + \sum_{\tau=0}^4 \gamma_{\tau} \times GDP_{t-\tau} + \kappa \times Pop_t,$$

where $\{c_0, \gamma, \kappa\}$ are parameters to be calibrated.

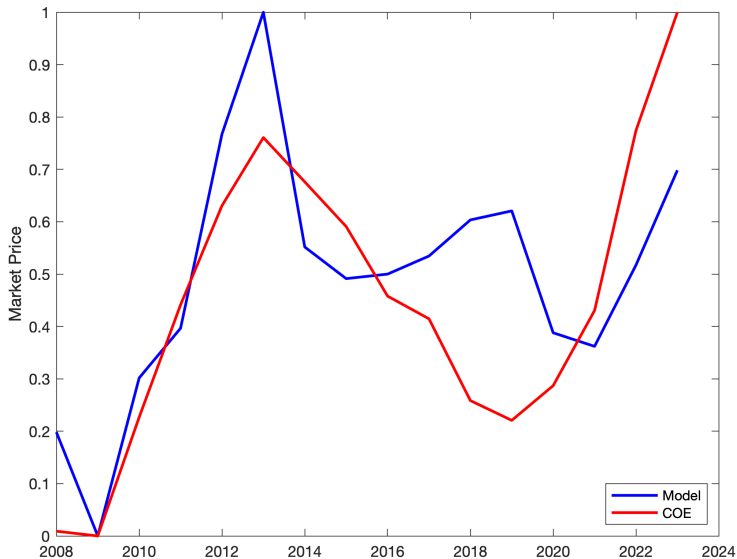
Calibration procedure

Calibrate parameters using Non-Linear Least Squares.

- 1 Given a guess of parameters, solve for market-clearing prices $\hat{p}_t$ in each year $t \in \{2007, 2008, \dots, 2023\}$. ▶ Market-Clearing Condition
- 2 Find parameters that minimise the squared distance between
 - The model-implied path of standardised COE prices, $\hat{p}_t$; and
 - The actual path of standardised COE prices, $COE_t(0)$.

▶ Formal statement of minimisation problem

Model Fit (Standardised prices relative to 2009)



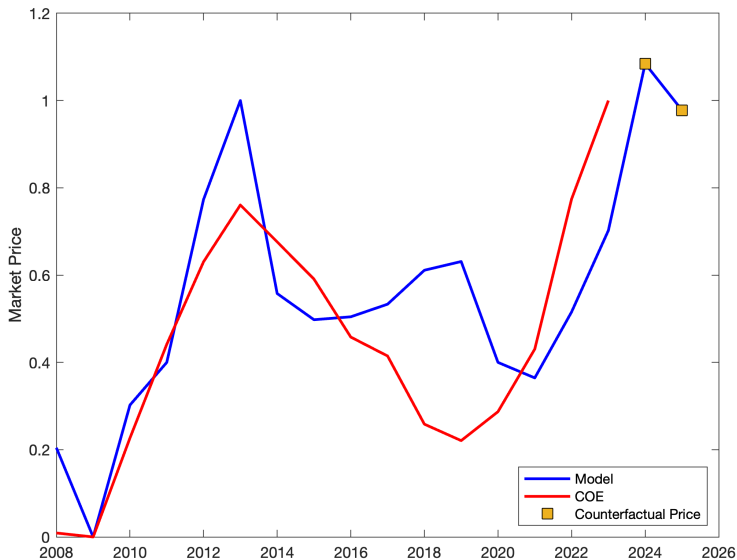
Suggestive Predictions

Calibrated model can predict 2024/2025 values of

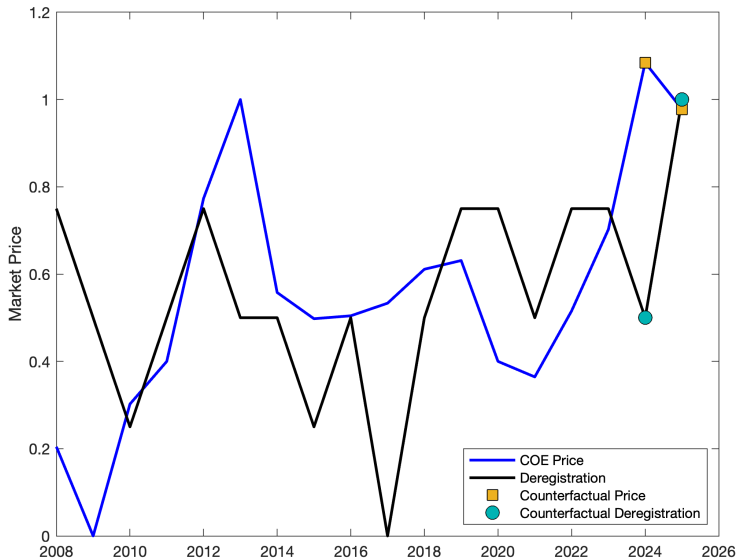
- COE prices;
- Replacement rates;
- Government revenue from VQP; and
- Welfare.

Today: focus on COE prices and deregistration rates.

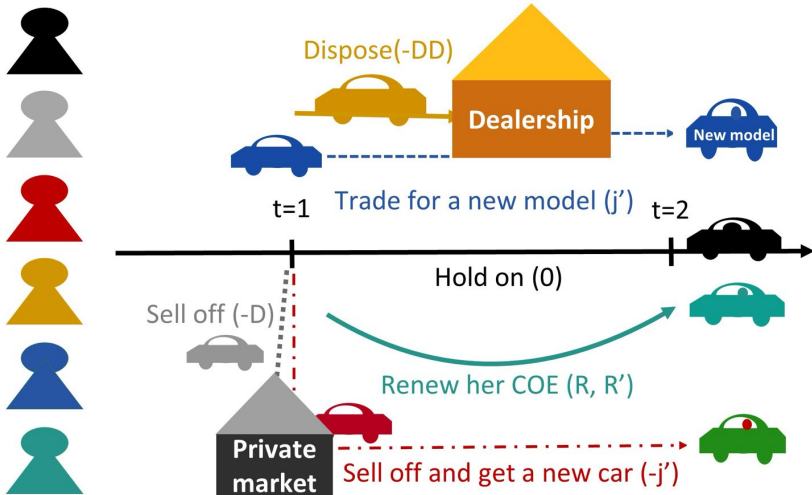
Suggestive Predictions: Graph



Suggestive Predictions: Graph



Preview of Full Model



LTA microdata informs consumer heterogeneity

Where would LTA microdata come in? Enrich the model by allowing

- Consumers to differ by residence and income type, proxied by 2-digit postal code and address type.
- Cars to differ on observables (make/model/year).

Every period (year),

- if a consumer has no car, she decides whether to buy it (and which);
- If she has a car, she decides whether to hold/sell/scrap; if car sold/scrapped, whether to buy a new one (and which).
 - If her COE is one year away from expiry, she can also choose to renew the COE at the PQP, as well as how long to renew it for (5 or 10 years)

► Model Details

- New cars imported at world price (+ prevailing taxes).

Future directions

We can do much more with microdata.

- Formulate descriptive evidence: Who are the car owners? What do they buy and where do they live?
- Estimate model of car ownership and replacement
- Simulate COE price path forward: Cuts and fills, etc.

Role of behavioural frictions?

"If I buy my car at \$170k, my view of it is pegged to the price I paid. So I'm loath to sell off the car at a price of \$200k, but maybe I can be convinced if it 'appreciates' to \$300k."

– W.T., economist

Thank You for Your Attention!

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Cars are rationed by permits

A car is a durable good identified by age a and exogenous characteristics x . It is subject to a COE quota. COEs are “stuck” to the car and differentiated by age τ .

- Mass Q_t of COEs in circulation, evolving according to λ_Q
- New COEs valid for $\bar{\tau}$ periods. Assume $2|\bar{\tau}$
- COE expires/car deregistered? Permit returns to government to be allocated same period
- For each period t : new COE price $COE_t(0)$; old COEs scrappable at

$$COE_t(\tau) = g(\tau) \times COE_t(0)$$

for some non-increasing function $g : \mathbb{Z}^+ \rightarrow [0, 1]$.

Assortment of all cars available at period t : $\mathcal{J}_t$

Consumer characteristics

Mass M_t of consumers live forever and discount future utility by a factor $\delta \in (0, 1)$. A consumer i is identified by their type ζ_i , where ζ_i denotes their residence and income type (2-digit postal code + address type). Consumers have rational expectations over the evolution of market state variables, including COE prices.

- Whether they own a car (and which type). Max one car.
- Value of car ownership:

$$u_{it}(\underbrace{a, \tau, x}_{\omega}) \sim F_{it}(a, \tau, x) = F(a, \tau, x, \zeta_i, s_t, \nu_{it}),$$

includes also aggregate states s_t and idiosyncratic shocks ν_{it}

At beginning of t , all consumers with $\bar{\tau}$ -period old cars must scrap them, thus becoming non-owners.

Non-owner problem

If a consumer does not have a car, every period, they look at the state of the market (including COE prices) and decide whether or not to purchase a car. If they buy a car, they must choose what type of car to buy. The value function of a consumer i without a car is

$$V_{it}(\emptyset) = \max_{d \in \mathcal{J}_t \cup \{0\}} v_{idt}(\emptyset) + \epsilon_{idt}.$$

If she chooses to remain without a car, her choice-specific value is

$$v_{i0t}(\emptyset) = \delta E_t V_{t+1}(\emptyset).$$

She obtains flow utility zero from not owning a car today, then considers her expected value of not owning a car next year, $E_t V_{t+1}(\emptyset)$, discounted by $\delta \in (0, 1)$.

Choice-specific values: No car $\rightarrow$ Car

If she chooses to acquire a car j :

$$v_{ijt}(\emptyset) = -\alpha_i p_{jt} + u_{it}(\omega_{jt}) + \delta E_t V_{t+1}(\omega_{jt+1}).$$

The consumer likes better flow utility $u_{it}(\omega_{jt})$ from car characteristics and a cheaper car p_{jt} . Continuation values are discounted by $\delta \in (0, 1)$.

Consumers with a car

If a consumer has a car j , of state (age, COE age, characteristics) $\omega = (a, \tau, x)$, she may take one of five types of actions to solve her dynamic programme:

$$V_{it}(a; \omega) = \max_{d \in \mathcal{J}_t \cup -\mathcal{J}_t \cup \{0, DD, D, R, R'\}} v_{idt}(\omega) + \epsilon_{idt}.$$

- ① Hold on to her car (action 0);
- ② Sell off her car on the private market (action D);
- ③ Sell off her car on the private market then buy another (action $-j'$);
- ④ Dispose of her car at a dealership (action DD);
- ⑤ Trade her car in to a dealer for a different model (action j');
- ⑥ Renew her COE for (half) the full duration (actions R, R').

Choice-specific values: Holding on to a car

If a consumer holds ($d = 0$), she enjoys the flow utility of car ownership and continues to the next period:

$$v_{i0t}(\omega) = u_{it}(\omega) + \delta E_t V_{it+1}(\omega').$$

Choice-specific values: Selling cars to the private market

If a consumer chooses to sell off her car on the private market ($d = D$), then she obtains the secondary market value of her car and continues to the next period car-less:

$$v_{iDt}(\omega) = \alpha_i p_t(\omega) + \delta E_t V_{it+1}(\emptyset)$$

If a consumer sells off her car on the private market, then replaces it with a car of variety j' ($d = -j'$), she enjoys the flow utility of car ownership net of the associated payments, then continues to the next period:

$$v_{i,-j',t}(\omega) = \alpha_i [p_t(\omega) - p_{j't}] + u_{it}(\omega_{j't}) + \delta E_t V_{it+1}(\omega_{j't+1}).$$

Choice-specific values: Disposing of cars at a dealership

If a consumer disposes of her car at a dealership ($d = DD$), then she obtains the pro-rated quota premium plus a scrap value, then continues to the next period car-less:

$$v_{i,DD,t}(\omega) = \alpha_i \check{p}_t(\omega) + \delta E_t V_{it+1}(\emptyset)$$

If she trades her car in for a model of variety j' ($d = j'$), she obtains the pro-rated quota premium plus a scrap value, enjoys the flow utility from owning her new car, then continues to the next period:

$$v_{ij',t}(\omega) = \alpha_i [\check{p}_t(\omega) - p_{j',t}] + u_{it}(\omega_{j',t}) + \delta E_t V_{it+1}(\omega_{j',t+1})$$

Choice-specific values: Renewing COEs

If she renews her COE for (half or) the full duration ($d \in \{R, R'\}$), she pays (half) the prevailing COE price, enjoys the flow utility of car ownership, then continues to the next period:

$$v_{iRt}(\omega) = \alpha_i[0.5 \times COE_t(0)] + u_{it}(a, \bar{\tau}/2, x) + \delta E_t V_{it+1}(a+1, \bar{\tau}/2+1, x);$$
$$v_{iR't}(\omega) = \alpha_i COE_t(0) + u_{it}(a, 0, x) + \delta E_t V_{it+1}(a+1, 1, x).$$

New car prices are world prices plus COE

On the supply side, we assume that the (permit-exclusive) prices and characteristics of new cars are set exogenously. As Singapore is a small, open economy in which all cars are imported, as a first pass we will assume that local dealers have no market power.

Given this assumption on supply, the price a consumer pays for her new car is the world price $\tilde{p}_{jt}$ plus the price of the permit:

$$p_{jt} = p_t(0, 0, x_{jt}) = \tilde{p}_{jt} + COE_t(0).$$

Meanwhile, prices of used cars set their demand equal to their supply.

An equilibrium in this economy comprises a sequence of allocations $(\mu_t)_{t \geq 0}$ of cars by age and consumer identities, a path of permit prices $(COE_t(0))_{t \geq 0}$, and a path of new and used car prices $(p_{jt})_{j \in \mathcal{J}_t, t \geq 0}$ such that

- ① *Consumers optimise.* Given the path of aggregate states, permit prices, and car prices, the allocations $(\mu_t)_{t \geq 0}$ are consistent with consumer optimality.
- ② *Prices clear markets.* Given the path of aggregate states and realised allocations $(\mu_t)_{t \geq 0}$, permit and car prices clear their respective markets.

COE market-clearing condition

For each period t , the COE price path p satisfies

$$\begin{aligned} \sum_{j:\tau_{jt}=0} \left[\int s_{jt}(\emptyset, p) d\mu_t(\emptyset, p) + \int [s_{jt}(\omega, p) + s_{-jt}(\omega, p)] d\mu_t(\omega, p) \right] + \\ \int s_{Rt}(\omega, p) + s_{R't}(\omega, p) d\mu_t(\omega, p) = \\ \int s_{DD,t}(\omega, p) + s_{0(t-1)}(\omega, p) + \sum_{j':\tau_{j't}=0} [s_{j't}(\omega, p)] d\mu_t(\omega, p). \end{aligned} \quad (1)$$

- First two lines: COE demand.
 - ① (Non-)Owners who want new cars.
 - ② Car owners who renew their COEs.
- Third line: COE supply. Three sources:
 - ① Voluntary early deregistrations (without replacement);
 - ② Mandatory scrappages;
 - ③ Car owners who trade in their cars to the dealer for new cars.

Objective function for calibration procedure

The calibration procedure solves the problem

$$\hat{\theta} = \arg \min_{\theta} \mathbf{m}(\theta)' \mathbf{m}(\theta)$$

such that predicted standardised COE prices are close to actual standardised COE prices:

$$\mathbf{m}(\theta) = \mathbb{E} \left[\frac{p_t - \min_t p_t}{\max_t p_t - \min_t p_t} - \frac{COE_t(0) - \min_t COE_t(0)}{\max_t COE_t(0) - \min_t COE_t(0)} \right]$$

► [Back to calibration procedure](#)

Data variables requested

- WR126: Rebate Issue Date, Rebate No., Rebate Type (PARF/COE), Rebate Owner ID, Rebate Owner ID Type, Rebate Expiry Date, Rebate Transfer Date, Rebate Used Date, Used On, Rebate Amount, COE Status Code, TCOE Issue Date, COE No., Bid Catg. Code, Reg. Catg. Code, Owner ID, Owner ID Type, Third Party Bidder ID, TCOE Expiry Date, TCOE Transfer Date, TCOE Cancel Date, COE Issue Date, COE Start Date, COE Expiry Date, Bid Amount, QP Amount, LSB Amount, Dereg. Vehicle No., Dereg. Vehicle Chassis, Dereg. Vehicle Engine, Use on Vehicle Number
 - Vehicle chassis and owner IDs needed to follow car across time.
 - Make/model/year are correlated with owner willingness to pay for COE.
 - Bid dates, expiry dates, rebate types, etc. are decision-relevant for the car owner.
- OP901, etc: all except owner names (OP904). Owner postal codes to be provided at 2-digit level. Owner birth year (but not exact birthday).
 - Postal codes and address type are indicative of owner income, thus willingness to pay for COE.
 - Owner age is indicative of income, and also proxies for household size.

Additional data variables to merge in

- New car prices and characteristics from, e.g., sgcar mart
- Price indices of cost shifters (e.g., steel)
- Prices of cars in adjacent markets (e.g., Malaysia, Indonesia?)

▶ Back to Calibration

Take model to data to recover consumer preferences and demand

- Challenge: Owners are forward-looking when deciding whether to replace a car
 - Usual approach: Given guess of preferences, simulate car owner behaviour many times and check that behaviour matches data
 - What we will use: “Euler CCP” techniques to “cancel out” the future value of holding a car (e.g., Arcidiacono and Miller 2011)
- Challenge: Cars of higher (unobserved) quality sell at higher prices
 - Solution: Cost shifters (e.g., steel) move car prices without moving unobserved quality

Estimation: Intuition

Consider two consumers with the same income, both owning a Toyota Corolla.

- The first owner sells off her car today; the other sells tomorrow.
- Both owners ended up “at the same place”, i.e., they no longer own a car. The future “looks the same” to them.
- Comparing these two owners, we can “cancel out” the part that affects their future value of owning a car (or not).
 - Informs us of the **comparison** between holding vs. selling a car today.